

Non-Graded Project: Course-End
Capstone Project: The Victor
Bushell Investigation

1. Project Overview

Welcome to your final capstone project. In this scenario, you will act as a digital forensics investigator for a law enforcement agency. You have been tasked with examining a suspect's laptop to find evidence related to a serious criminal investigation. This project requires you to apply the legal, ethical, and technical skills from all four modules, using the Autopsy Digital Forensics Platform to analyze a real-world forensic image.

2. Prerequisites

This is a cumulative project. Before beginning, you must have a thorough understanding of all content from Modules 1 through 4, including all video scripts, hands-on labs, and discussion prompts.

3. Learning Objectives

This capstone project assesses your ability to:

- Module 1: Apply principles of evidence handling and documentation within a legal framework.
- Module 2: Use a digital forensics tool (Autopsy) to conduct a systematic examination of a disk image, ensuring evidence integrity.



- Module 3: Analyze and correlate different types of digital artifacts (in this case, web history and file downloads) to reconstruct user activity and identify potential evidence.
- Module 4: Synthesize technical findings into a clear, objective, and defensible forensic report suitable for legal proceedings.



4. The Scenario: The Victor Bushell Case

Background: On October 19, 2011, a man named Victor Bushell was arrested at the Port of Dover, UK, while trying to travel to France. He was carrying a forged passport and a large amount of cash. Authorities suspect he is involved in the illegal importation of weapons into the United Kingdom.

Initial Triange: Mr. Bushell's laptop was seized at the time of his arrest. A forensic image of the laptop's hard drive has been created. The lead investigator has tasked you with a specific objective: examine the suspect's internet browsing history to find any evidence that supports the allegation.

Your Role: You are the digital forensics examiner. Your mission is to conduct an impartial and thorough examination of the provided disk image, document your process, and report your findings. You must adhere to the ethical and legal standards discussed throughout this course.

5. Provided Evidence & Tools

Primary Evidence: A compressed forensic disk image named 2011-10-19-Image.zip. This contains a bit-for-bit copy (an EnCase E01 file) of the hard drive from the suspect's computer.

Primary Tool: The Autopsy Digital Forensics Platform.

6. Your Tasks & Deliverables

You must produce a single document, `bushell_capstone_report.md`, that contains the following three sections.

Section 1: Case Setup & Evidence Handling (Modules 1 & 2)

Task 1: Install Autopsy: Download and install the latest version of Autopsy from <https://www.autopsy.com/>

Task 2: Download & Extract Evidence: Download the 2011-10-19-Image.zip file from <http://www.digital-detective.net/example/2011-10-19-Image.zip>. Unzip the file to access the 2011-10-19-Image.E01 evidence file within it.

Task 3: Create Case & Ingest Data: Create a new case in Autopsy named Bushell Investigation. Add the 2011-10-19-Image.E01 file as a data source. Configure and run all the standard Ingest Modules (such as Recent Activity, Keyword Search, etc.) to process the data.

Deliverable: In your report, briefly describe the steps you took to create the case and add the evidence, explaining why this structured approach is vital for maintaining the chain of custody and ensuring your findings are admissible.

Section 2: Forensic Analysis (Module 3)

Task: Once the ingest modules are complete, conduct a systematic review of the evidence within Autopsy to answer the following investigative questions. The primary area of focus is the suspect's web browsing activity.

Deliverable: For each question below, provide a detailed answer in your report, citing specific data points from Autopsy (e.g., URLs, dates, downloaded filenames). Include at least two screenshots to support your key findings.

- i. **Web History Analysis:** Using the 'Web History' section in Autopsy, identify any websites the suspect visited that could be related to the illegal importation of weapons, travel to France, or creating forged documents. List at least three suspicious URLs and the dates they were visited.
- ii. **File Downloads:** Using the 'Web Downloads' section, did the suspect download any files of interest? List any downloaded files that seem relevant to the investigation.

iii. Keyword Searches: Examine the 'Web Search' results. Did the suspect search for any terms related to the case? List any relevant search queries and the date they were performed.

Section 3: Final Report & Conclusion (Module 4)

Task: Synthesize your findings from Section 2 into a professional summary.

Deliverable: Write a concluding section in your report that summarizes your findings. Based only on the digital evidence you analyzed, state whether you found evidence to support the allegation that the suspect was researching or planning illegal activities. Your conclusion must be objective, factual, and based strictly on your analysis.

Instructions for Documenting and Sharing the Project for Peer Review

Document the Project



Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.



Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.